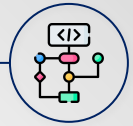
# Data structure and Algorithms

Presented by : Asmaa Ghonaim



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# Queue

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# Queue

- A Queue is a linear list of information that is accessed in First-In, First- Out (FIFO) order.
- The first item placed on the queue is the first item retrieved. The second item put in is the second item retrieved, and so on.
- A queue does not allow random access of any specific item.

**Example:**

Queues are very common in everyday life. For example, lines at a bank or a fast-food restaurant are queues.

Used in many types of programming situations such as simulations, event or appointment scheduling (such as in a PERT or Gantt chart), and I/O buffering.

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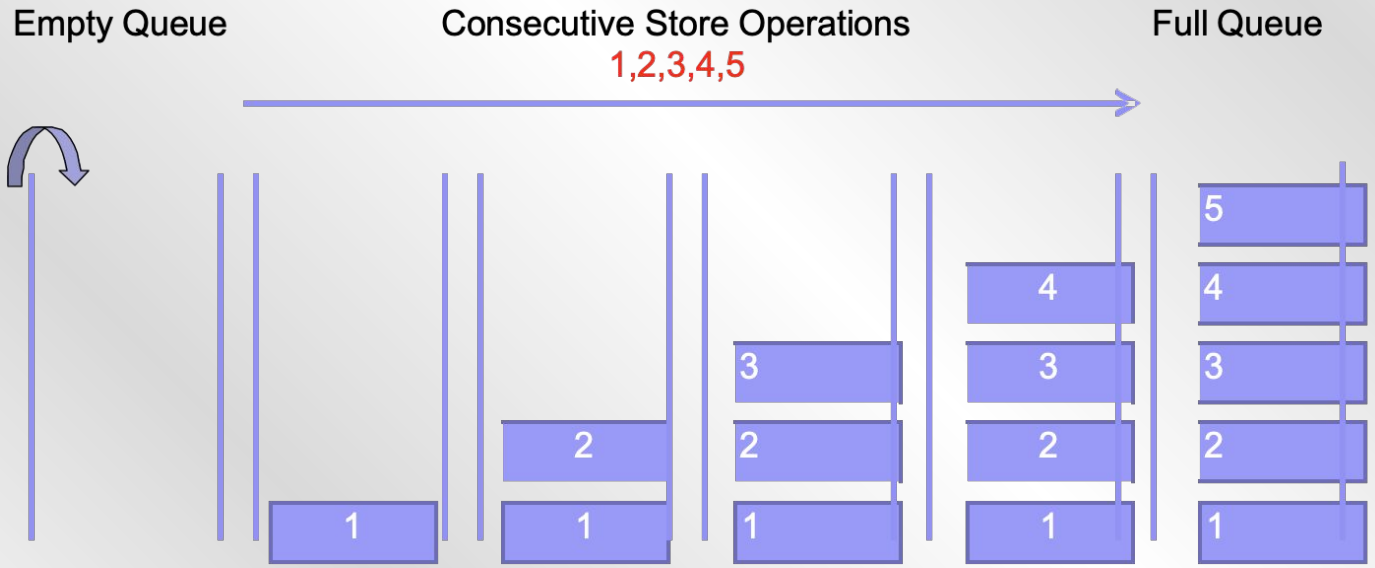
# Queue

To visualize how a queue works, consider-two-functions: **enqueue()** function places an item onto the end of the queue

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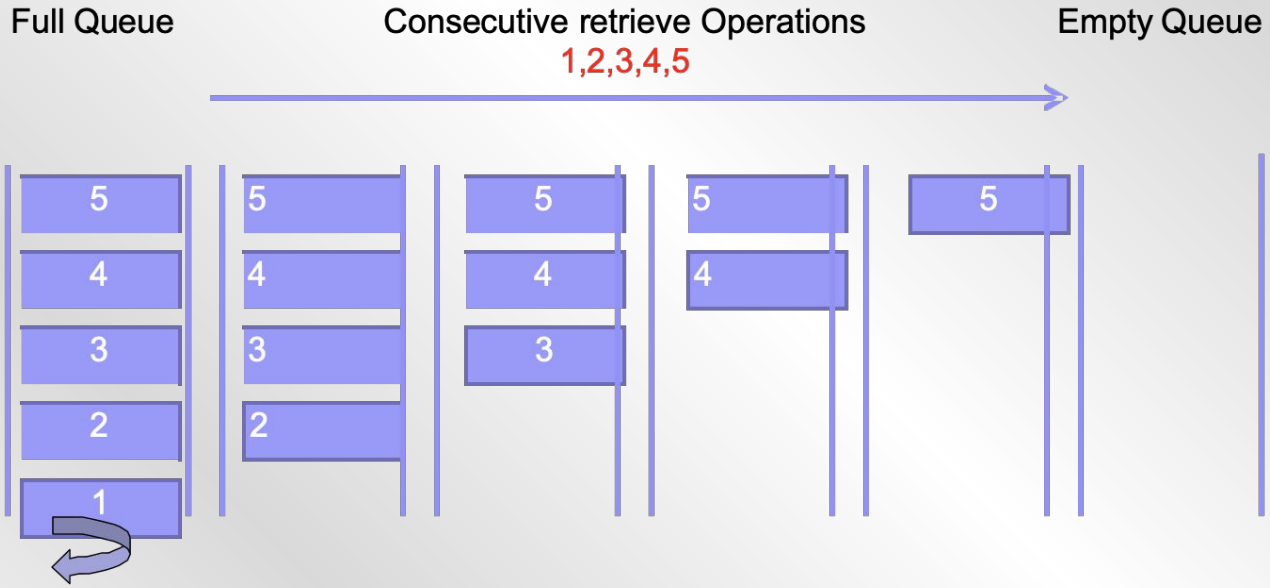
# Queue

To visualize how a queue works, consider-two-functions: **dequeue( )** function removes the first item from the queue and returns its  value, if the value is not stored elsewhere, destroys it.

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# Queue

## Queue implementation as a linked list for dynamic length:

### Analyze the Linked list class first;

```
class LinkedList {
protected:
Employee *pStart;  Employee *pEnd;
public:
LinkedList() { pStart=pEnd=NULL;}
~LinkedList() { freeList();}
// Setters and getters for pStart and pEnd;
 void addList(Employee *pItem);          / Needed : add to the end as the queue
 void InsertList(Employee *pItem);       // Not needed here
Employee* searchList(int Code);          // Needed
int deleteList(int Code);                // Needed :delete element from queue
void freeList();                         // Needed
void displayAll();                       // Needed
```

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# Queue

**Queue implementation as a linked list for dynamic length:**

Class Queue will inherits private from Linked List

// We need only some methods as the other methods are against the concept of the queue.

```
class Queue : private LinkedList {
        // No need for size, front, rear and Employee as the size is dynamic
public:
        Queue ( ): LinkedList() {

        } //No need for the constructor which take size
        ~Queue ( ) {
        }
        void enqueue (Employee *e){
                addList(e);                // or LinkedList :: addList(e);
        }
```

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# Queue

**Queue implementation as a linked list for dynamic length:**

```
Employee* dequeue (){
        Employee *pNode;
        pNode  = pStart;
        if(pStart){
                if(pStart == pEnd){        // there is only node
                        pStart = pEnd =NULL;
                }
                else {    // first elment
                        pStart = pStart -> pNext;
                        pStart -> pPrevious = NULL;
                }
        }
  return pNode;
}
```

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# Queue

**Queue implementation as a linked list for dynamic length:**

```
        Employee*  searchQueue(int Code){
                return   searchList(Code);
        }

        int DeleteQueue(int Code){
                return DeleteList(Code);
        }
        void freeQueue(){
                freeList();
        }
        void displayAll(){
                LinkedList::displayAll();
        }
};
```

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# Stack

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# Stack

A Stack is the opposite of a Queue because it uses Last-In, First-Out (LIFO) Accessing.

- This means elements are removed from the stack in the reverse order to the order of their addition: therefore, the lower elements are typically those that have been in the list the longest.
- Insert: **PUSH**
- Remove: **POP**
- The accessible element is called **TOP**.
- Like the queue, the retrieval function takes a value off the list and, if the value is not stored elsewhere, destroys it.

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# Stack

Example:

- Imagine a stack of plates. The bottom plate in the stack is the last to be used, and the top plate (the last plate placed on the stack) is the first to be used.
- Stacks are used a great deal in system software including compilers and interpreters. In fact, C uses the computer's stack when passing arguments in functions.

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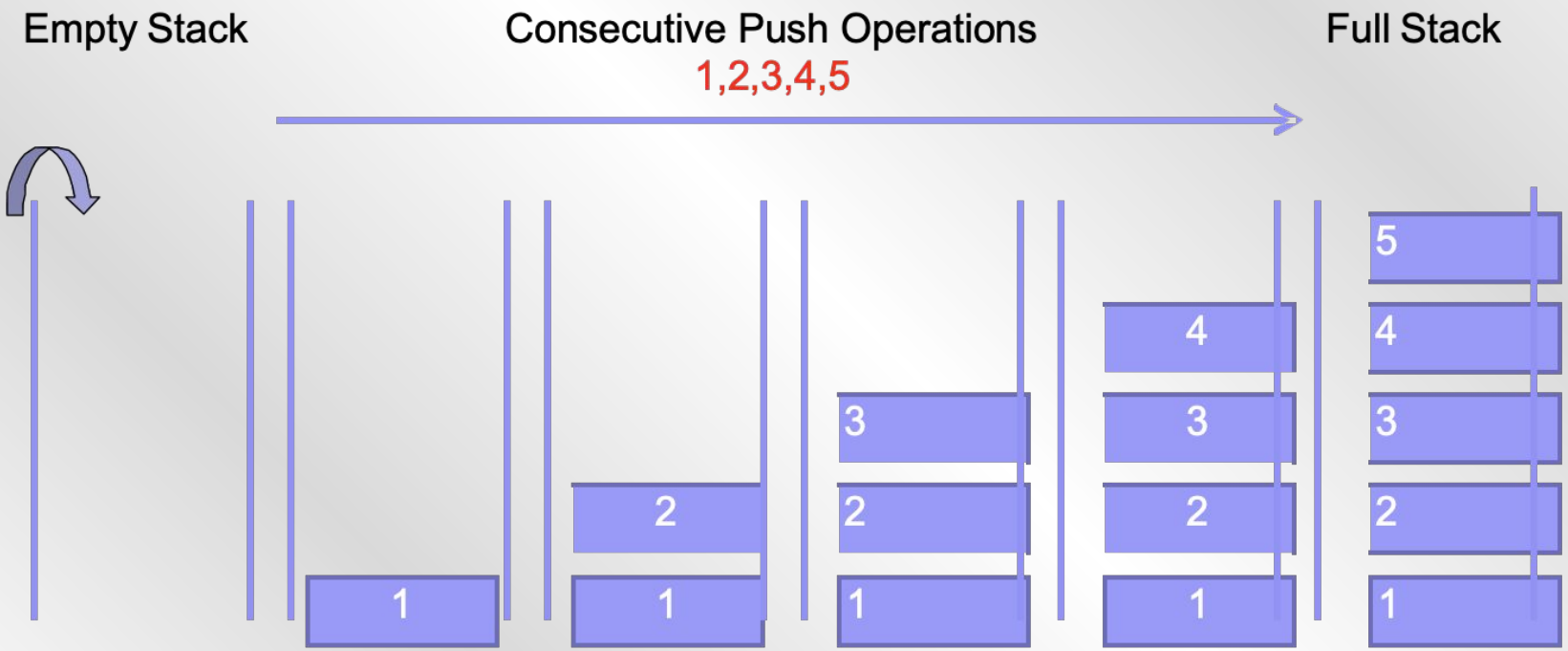
# Stack

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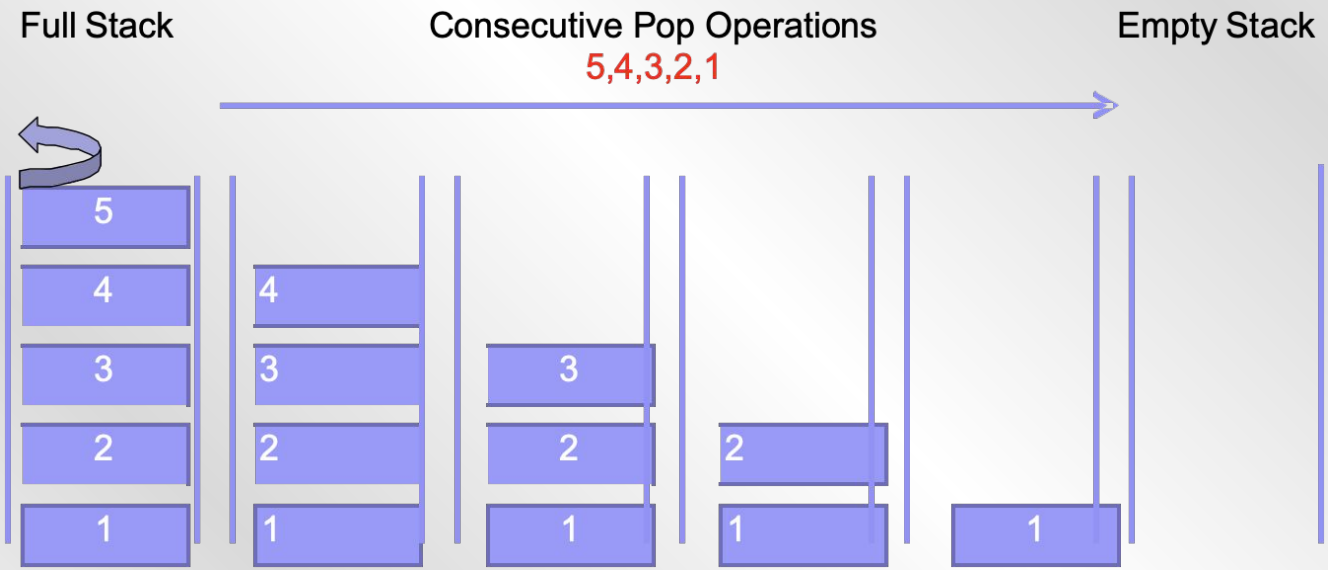
# Stack

To visualize how a Stack works, consider-two-functions:

- Pop (Retrieve) Operation : Pop operation retrieves a value from the stack.
-

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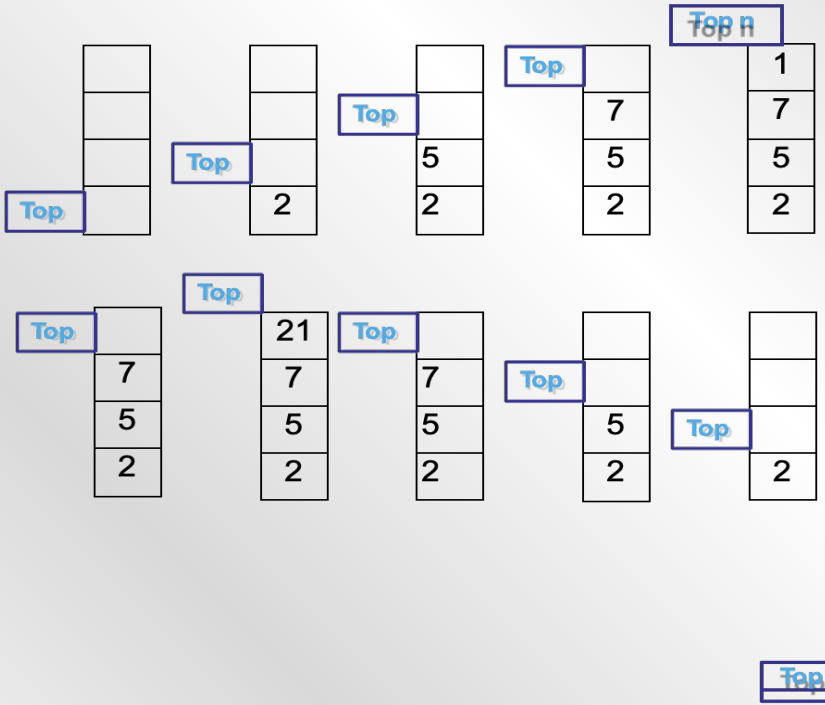
# Stack




- Push (2)
- Push (5)
- Push (7)
- Push (1)
- Push (3) // stack is full 3 not added
- Pop()→ 1
- Push (21)
- Pop()→21
- Pop()→7
- Pop()→5
- Pop()→2
- Pop()→ // stack is empty

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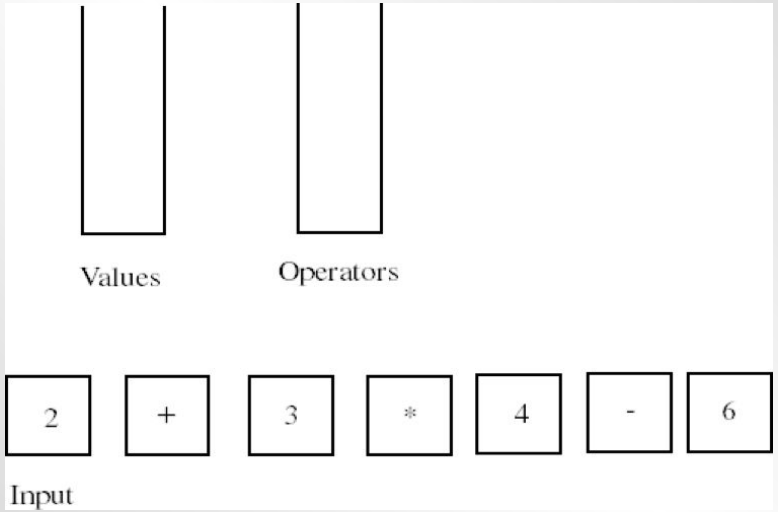
# Evaluating Arithmetic Expressions

**Operator precedence:**

- * has precedence over +/–
- operators of the same precedence group evaluated from left to right \

2 + 3 * 4 - 6

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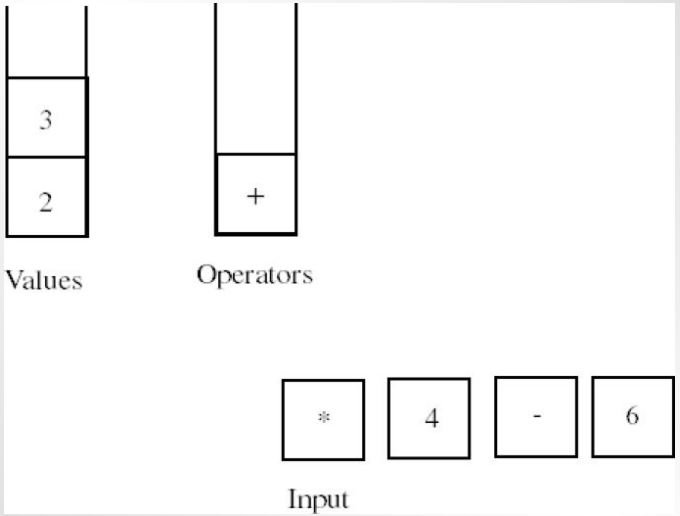
# Evaluating Arithmetic Expressions

- 2 + 3 * 4 – 6 = 8
- After handling the first three tokens the next operator in the input has a higher precedence than the operator at the top of the operator stack, so we push the next operator. The next number token also gets pushed on the value stack.

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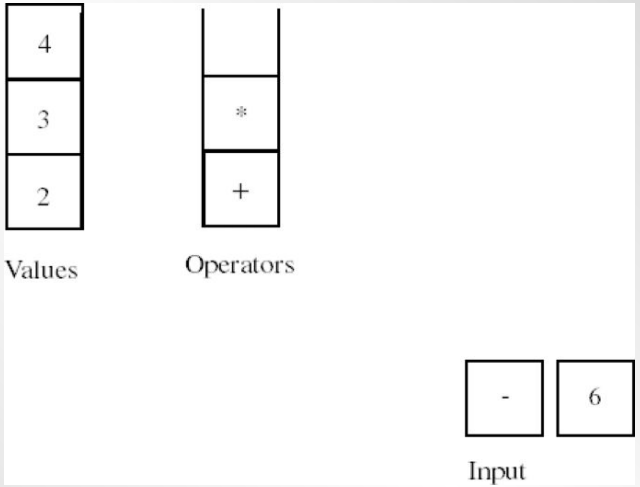
# Evaluating Arithmetic Expressions

- 2 + 3 * 4 – 6 = 8
- The next operator in the input sequence has a precedence lower than that of the operator at the top of the operator stack. This causes us to process and remove the operator at the top of the operator stack. **3 * 4**

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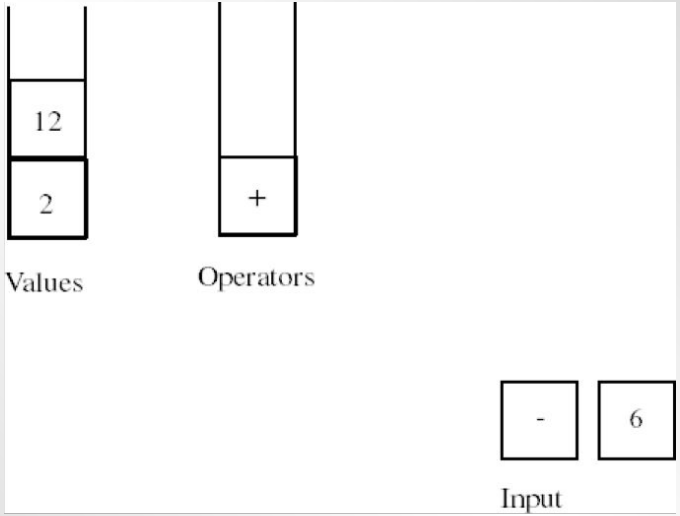
# Evaluating Arithmetic Expressions

- 2 + 3 * 4 – 6 = 8
- Once again, the operator in the input has a precedence equal to that of the operator at the top of the operator stack, so we process and remove the operator from the operator stack. **2 + 12**

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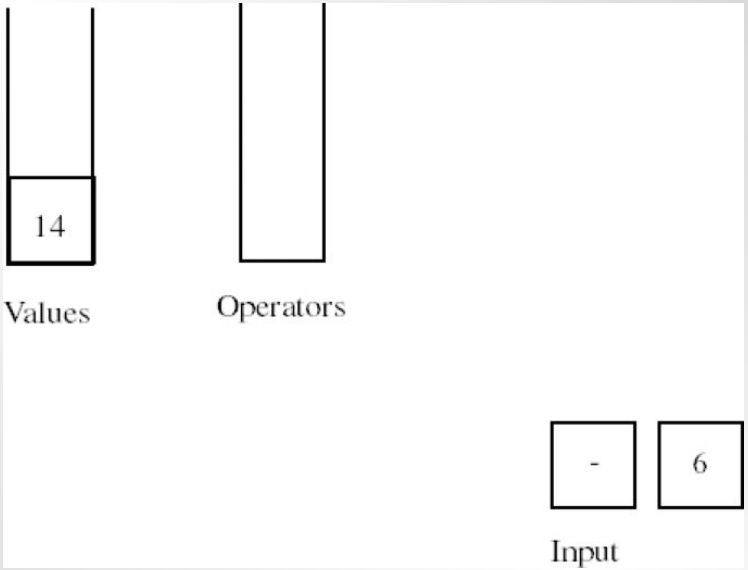
# Evaluating Arithmetic Expressions

- 2 + 3 * 4 – 6 = 8
- At this point the operator stack is empty, so the operator  token at the front of the input gets pushed on the operator  stack. The number token at the end of the input gets pushed on the value stack.

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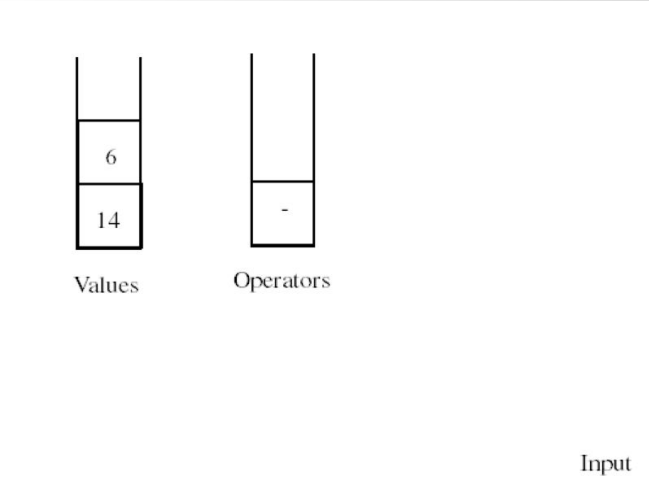
# Evaluating Arithmetic Expressions

- 2 + 3 * 4 – 6 = 8
- Once the input has emptied out, we process any operators that remain on the operator stack. Once all of those operators have been processed, the sole remaining number on the value stack is the result of the computation. **14 - 6**

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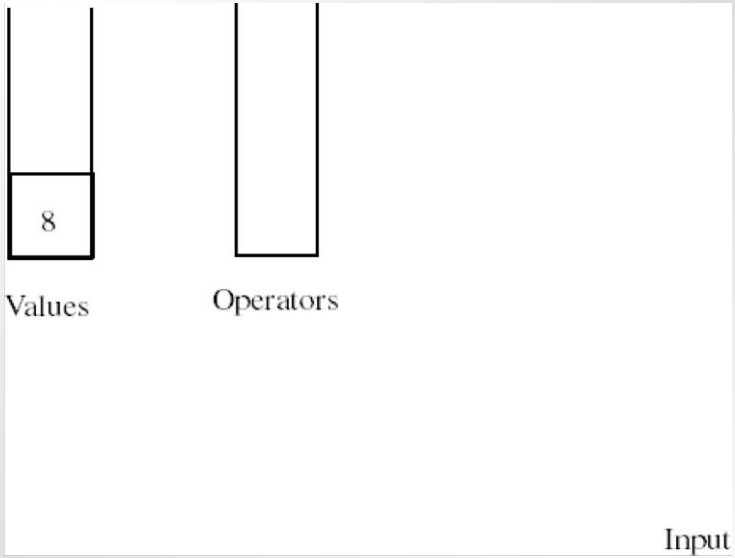
# Evaluating Arithmetic Expressions

- 2 + 3 * 4 – 6 = 8




Try **2 + 4 -5 * 10 / 2 +1 ?**

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# Evaluating Arithmetic Expressions

**The basic algorithm:**

1. Make use of two TokenStacks [ a *value stack* ,an *operator stack]*.
2. If character is operand push on the operand stack.
3. Else if character is operator
    1. While the top of the operator stack is not of smaller  precedence than this character.
    2. **Pop** operator from operator stack.
    3. **Pop** two operands (op1 and op2) from operand stack.
    4. Store op1 op op2 on the operand stack back to 2.1.
4. Else (no more character left to read):
5. **Pop** operators until operator stack is not empty.
6. **Pop** top 2 operands and push op1 op op2 on the operand stack.
7. return the top value from operand stack.

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# Lab
# Exercise

# Assignments :

- ➔ Use Stack to evaluate this operation **3 + 4 / 2 + 2 * 12 - 6**
- ➔ Implement stack using linked list

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